

ISLE OF MAN, Isle of Man

WMO: 032040

Lat: 54.0853N

Lon: 4.6319W

Elev: 17

StdP: 101.12

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	0.9	-5.5	2.4	2.3	-4.4	2.6	2.5	18.6	7.8	16.9	7.2	4.3	10	0.739

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	20.9	16.1	19.2	15.3	18.0	14.7	16.9	19.5	16.1	18.0	15.6	17.2	4.7	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.3	17.5	15.3	10.9	16.9	14.8	10.5	16.4	47.3	19.7	45.2	18.1	43.6	17.2	19.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	14.2	12.7	DB	-2.3	23.6	1.4	2.1	-3.3	25.1	-4.1	26.4	-4.9	27.6	-5.9	29.1	(4)
(5)			WB	-3.5	18.3	1.4	1.1	-4.4	19.1	-5.2	19.7	-6.0	20.3	-7.0	21.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	6.3	6.0	6.7	8.3	10.9	13.2	15.0	15.2	13.8	11.5	8.9	7.0
	DBStd	3.93	2.37	2.25	2.10	2.02	2.19	1.87	1.73	1.45	1.57	2.13	2.49	2.70
	HDD10.0	554	114	112	103	56	14	1	0	0	1	11	48	95
	HDD18.3	2948	372	345	361	300	231	153	104	100	137	211	284	351
	CDD10.0	651	1	0	1	6	42	97	156	160	114	58	14	3
	CDD18.3	4	0	0	0	0	0	1	2	1	0	0	0	0
	CDH23.3	12	0	0	0	0	0	2	7	3	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.3	7.9	7.4	6.7	5.7	5.5	5.0	4.7	5.0	5.7	7.0	7.2	7.8
Precipitation	PrecAvg	956	94	73	65	60	62	65	63	70	75	112	111	106
	PrecMax	1264	193	136	181	135	138	195	98	156	162	254	230	208
	PrecMin	725	16	19	22	15	16	17	28	7	4	41	59	59
	PrecStd	134	41	33	32	27	32	38	20	36	36	53	42	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	11.1	12.2	16.2	20.6	22.2	24.3	22.7	19.1	16.3	14.2	12.7
		MCWB	10.3	9.5	9.2	11.2	14.2	15.5	17.5	17.2	15.9	14.2	13.1	11.5
	2%	DB	10.9	10.2	11.0	13.7	17.7	19.5	21.3	19.9	17.7	15.6	13.5	12.0
		MCWB	9.7	9.0	8.8	9.8	13.3	14.9	16.6	16.2	14.9	13.9	12.4	11.0
	5%	DB	10.2	9.7	10.2	12.2	15.8	17.6	19.2	18.5	16.9	15.0	12.9	11.4
		MCWB	9.1	8.6	8.4	9.2	12.2	13.8	15.4	15.3	14.5	13.6	11.7	10.3
	10%	DB	9.7	9.2	9.5	11.2	14.2	16.3	17.8	17.5	16.2	14.5	12.3	10.8
		MCWB	8.6	8.0	8.0	8.8	11.3	13.2	14.7	14.8	14.1	13.1	11.0	9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.1	10.2	11.6	15.2	16.8	18.2	18.0	16.7	15.1	13.6	12.0
		MCDB	11.2	10.5	11.2	15.0	19.0	20.3	22.5	21.3	18.1	15.6	14.0	12.3
	2%	WB	10.1	9.5	9.5	10.5	13.7	15.4	17.0	16.8	15.8	14.5	12.7	11.3
		MCDB	10.6	10.0	10.4	12.6	17.0	18.4	20.3	18.8	16.7	15.2	13.3	11.8
	5%	WB	9.5	8.9	8.9	9.9	12.7	14.5	16.1	16.1	15.3	14.0	12.0	10.6
		MCDB	10.2	9.6	9.7	11.5	15.1	16.9	18.4	17.6	16.3	14.9	12.7	11.3
	10%	WB	8.9	8.3	8.4	9.3	11.8	13.7	15.4	15.6	14.7	13.4	11.3	9.8
		MCDB	9.6	9.1	9.3	10.8	13.7	15.7	17.2	16.9	15.9	14.4	12.2	10.7
Mean Daily Temperature Range	5% DB	MDBR	3.5	3.8	4.3	4.9	5.4	5.1	5.0	4.8	4.5	4.0	3.8	3.7
		MCDBR	3.7	3.4	4.4	6.1	7.6	7.3	7.3	6.1	5.0	3.9	3.6	3.8
		MCWBR	4.2	3.5	3.5	4.0	4.6	4.2	4.1	3.8	3.7	3.7	4.2	4.4
	5% WB	MCDBR	3.7	3.3	3.9	5.0	6.6	6.3	6.1	5.0	4.3	3.5	3.5	3.7
MCWBR		4.3	3.7	3.7	3.6	4.4	4.0	3.9	3.6	3.7	3.8	4.2	4.4	
Clear-Sky Solar Irradiance	taub	0.333	0.344	0.370	0.388	0.392	0.386	0.387	0.373	0.366	0.351	0.328	0.320	
	taud	2.292	2.372	2.341	2.328	2.374	2.423	2.421	2.474	2.488	2.457	2.365	2.268	
	Ebn at Noon	606	744	803	839	855	864	855	848	807	730	615	536	
	Edh at Noon	60	77	97	112	112	108	107	96	84	70	56	51	
All-Sky Solar Radiation	RadAvg	0.59	1.29	2.50	4.01	5.13	5.28	4.95	4.12	2.88	1.58	0.75	0.43	
	RadStd	0.06	0.12	0.19	0.28	0.40	0.48	0.41	0.29	0.18	0.16	0.08	0.04	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.18	+0.43	+0.75	N/A	N/A	N/A	N/A	-75	+45	N/A	
(47) Regional (51 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page